

2. A 120×120 pixel snapshot of the detected object is saved as `/snapshot/snapshot.jpg`.

3. When no smoker is detected, GPIO16 is activated, lighting a green LED to indicate that all is clear.

To retrieve `snapshot.jpg`, SCP can be used, or the file may be accessed through the MaixVision IDE.

Real-life application

This system operates in real time for smoker detection. It uses two GPIO pins—pin 15 and pin 16—to control external devices. One pin is configured for a buzzer and the other for an indicator.

Green LED. It lights up when no



Fig. 3: With cigarette



Fig. 4: Without cigarette



Fig. 5: Snapshot of the smoker

smoker is detected

Red LED and buzzer. Get activated to indicate and sound an alarm when a smoker is detected

The system captures a snapshot of the smoker and stores it on the SD card along with metadata. Three types of metadata are generated: date and time, detection confidence level, and bounding box details. The most relevant—date and time—is printed directly on the image. The other two are currently commented out in the code but may be enabled, if required.

Connecting and setting up the MaixCam

To begin, download the latest MaixCam operating system from the release page: <https://github.com/sipeed/MaixPy/releases>. In most cases, the board comes with an operating system pre-installed. If so, only a connection to a computer is needed. There are three main ways to connect the board to a computer:

1. USB direct connection
2. Through SSH
3. Using MaixVision IDE (<https://>

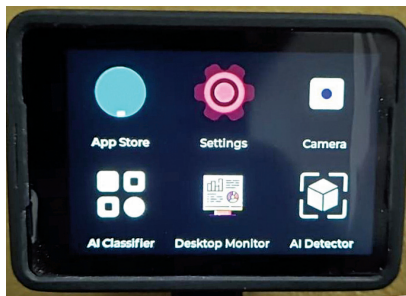


Fig. 6: On-screen instruction

wiki.sipeed.com/maixvision/) for a more visual and integrated experience

For uploading the OS

Either open the board's back cover, remove the SD card, upload the OS, and then reinsert the SD card, or follow the alternative method without opening the back cover: Press the user button, then insert the USB cord into the computer. The user button may need to be pressed a few more times until MaixCam appears as a USB drive labelled 'boot' in 'This PC.' Use 'balenaEtcher' (<https://www.balena.io/etcher/> or <https://github.com/balena-io/etcher>) or any similar software to transfer the OS image to the MaixCam drive.

Once the OS has been transferred, unplug the USB cord and power up again. The MaixCam will display the opening screen on the TFT display. Tap on 'Settings,' navigate to 'Wi-Fi,' scan for available networks, and connect using the Wi-Fi password.

After connecting to Wi-Fi, MaixCam may prompt a download of the MaixPy runtime. Follow the on-screen instructions to complete the setup (see Fig. 6). Once complete, the icons on the TFT display may be explored to better understand the board's features.

MaixVision IDE and SSH setup

MaixVision IDE appears on the left side of the screen, while the right side allows access to the MaixCam via SSH. From MaixVision, either SSH or GFTP (Graphical FTP) can be used to communicate with the board. Alternatively, SSH access may be gained directly through a computer's command prompt using: `ssh root@192.168.82.112` (get the IP address from the router). When prompted for a password, enter 'root'. Many sample codes are available at <https://maixhub.com/app/>.